

8.3 Rotation



Core skills you'll master in this section:

- ▶ Using matrix multiplication to implement rotation about the origin in the plane.
- ▶ The rotation matrix is an orthogonal matrix with determinant 1.
- ▶ Successive planar rotations satisfy the commutative law.
- ▶ The relationship between the Gram matrix and vector inner products.
- ▶ Swapping the order of rotation and scaling changes the result – matrix multiplication generally does not satisfy the commutative law.
- ▶ Using matrix multiplication to implement 3D rotation about an axis.

Rotation transformations are widely used in fields such as machine learning algorithms, computer graphics, robotics, and physics. Later in the book, you'll see rotation applied in topics such as spectral decomposition, principal component analysis, and singular value decomposition. Rotation is typically carried out through matrix multiplication. This section covers planar rotation first, then 3D rotation.

Planar Rotation

The transformation matrix for rotating the plane counterclockwise about the origin by angle θ is

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

In the plane, given a column vector $\mathbf{x}$, its new coordinates after rotating counterclockwise about the origin by angle θ are

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

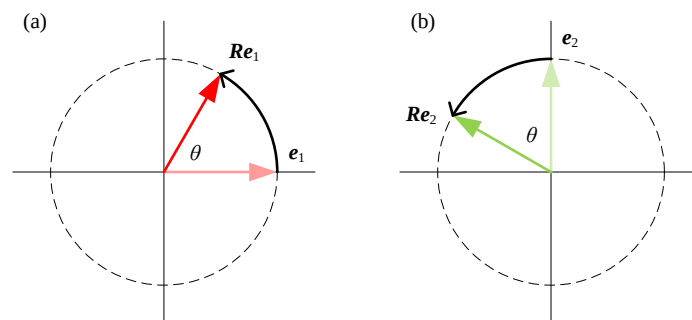


Figure 1. $\mathbf{R}$'s Action on $\mathbf{e}_1$ and $\mathbf{e}_2$

R 's action on $\mathbf{e}_1$ gives

$$R\mathbf{e}_1 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

The expression above is equivalent to extracting R 's first column vector.

As shown in Figure 1 (a), $\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ is $\mathbf{e}_1$'s new position after rotating counterclockwise about the origin by θ .

R 's action on $\mathbf{e}_2$

$$R\mathbf{e}_2 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

The expression above is equivalent to extracting R 's second column vector.

As shown in Figure 1 (b), $\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$ is $\mathbf{e}_2$'s new position after rotating counterclockwise about the origin by θ .

Figure 2 gives two examples.

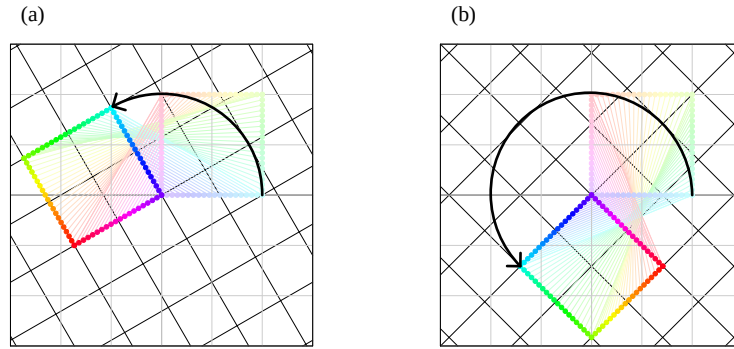


Figure 2. Two Examples of Rotation

Figure 2 (a) corresponds to the rotation matrix

$$R = \begin{bmatrix} \cos \frac{2\pi}{3} & -\sin \frac{2\pi}{3} \\ \sin \frac{2\pi}{3} & \cos \frac{2\pi}{3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

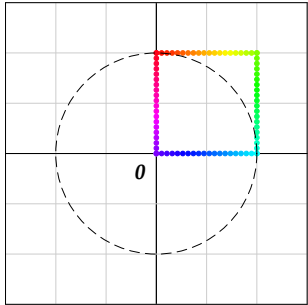
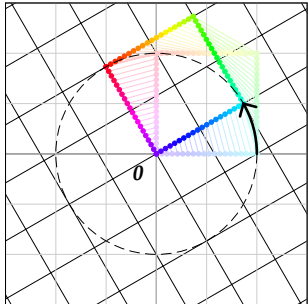
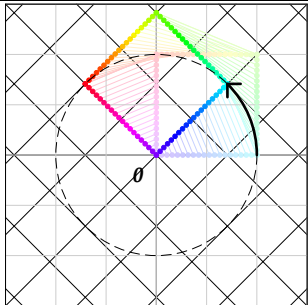
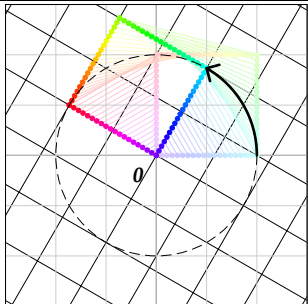
Figure 2 (b) corresponds to the rotation matrix

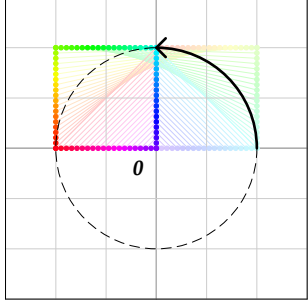
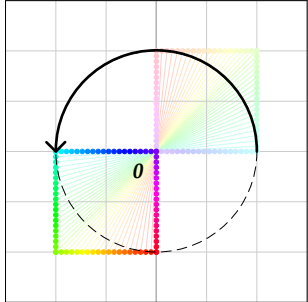
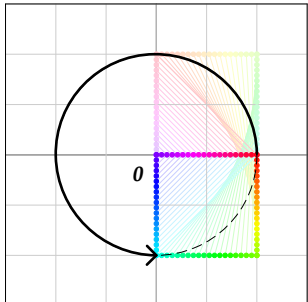
$$R = \begin{bmatrix} \cos\left(-\frac{3\pi}{4}\right) & -\sin\left(-\frac{3\pi}{4}\right) \\ \sin\left(-\frac{3\pi}{4}\right) & \cos\left(-\frac{3\pi}{4}\right) \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

Table 1 shows the rotation matrices and shape transformations for several common rotation angles.

? Please analyze each rotation matrix in Table 1: compute its determinant, compute its inverse, and apply it to $\mathbf{e}_1$ and $\mathbf{e}_2$.

Table 1. Several Different Rotation Angles

Counterclockwise Rotation	Matrix	Diagram
0° Equivalent to no change	$\begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Identity matrix	
30°	$\begin{bmatrix} \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$	
45°	$\begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$	
60°	$\begin{bmatrix} \cos \frac{\pi}{3} & -\sin \frac{\pi}{3} \\ \sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$	

90°	$\begin{bmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	
180°	$\begin{bmatrix} \cos \pi & -\sin \pi \\ \sin \pi & \cos \pi \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	
270°	$\begin{bmatrix} \cos \frac{3\pi}{2} & -\sin \frac{3\pi}{2} \\ \sin \frac{3\pi}{2} & \cos \frac{3\pi}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	

Inverse Matrix: The Inverse Operation

Taking the inverse of the rotation matrix $R(\theta)$

$$R^{-1}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

Figure 2 shows rotation and its inverse operation. It's easy to see that actually represents rotating clockwise about the origin by θ degrees.

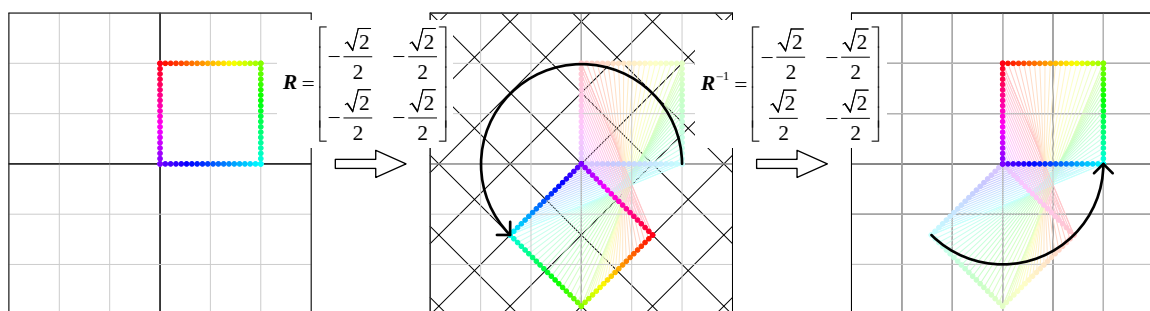


Figure 3. The Inverse of Rotation

Clockwise rotation by θ corresponds to counterclockwise rotation by $-\theta$; the corresponding rotation matrix is

$$\begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

and are the same matrix.

Determinant

The determinant of the 2D rotation matrix is

$$\det(\mathbf{R}(\theta)) = \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \cos^2 \theta + \sin^2 \theta = 1$$

This shows that planar rotation doesn't change a shape's area; this rotation (about the origin) also doesn't change the ordering of vectors.

Orthogonal Matrix: Orthonormal Basis

Comparing and , we find that the rotation matrix's inverse $\mathbf{R}^{-1}$ is its transpose $\mathbf{R}^T$;

$$\mathbf{R}^{-1} = \mathbf{R}^T$$

That is to say

$$\mathbf{R} @ \mathbf{R}^T = \mathbf{R}^T @ \mathbf{R} = \mathbf{I}$$

This means $\mathbf{R}$ is an orthogonal matrix.

Writing the rotation matrix $\mathbf{R}$ out in terms of its column vectors gives the following form

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} = [\mathbf{r}_1 \quad \mathbf{r}_2]$$

We find that each column of the rotation matrix is a unit vector (a direction vector); the square of its L^2 vector norm (vector length, vector magnitude, Euclidean norm) equals 1, that is

$$\begin{aligned} \|\mathbf{r}_1\|_2^2 &= \mathbf{r}_1^T @ \mathbf{r}_1 = 1 \\ \|\mathbf{r}_2\|_2^2 &= \mathbf{r}_2^T @ \mathbf{r}_2 = 1 \end{aligned}$$



Please compute on your own.



Please review the earlier material in this book on vector norms, matrix multiplication, and inner products.

Also, 's column vectors are orthogonal, that is

$$\mathbf{r}_2^T @ \mathbf{r}_1 = \mathbf{r}_1^T @ \mathbf{r}_2 = 0$$

This means that $\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2]$ is an orthonormal basis. For example, in Table 1, the grids for the orthonormal bases are all unit squares.

In particular, when $\theta = 0$, $\mathbf{R} = \mathbf{I}$, that is

$$\begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [\mathbf{e}_1 \quad \mathbf{e}_2]$$

In this case, what we get is the standard basis (also called the natural basis).



Please review the earlier material in this book on bases, orthonormal bases, and the standard basis.

Gram Matrix

With $\mathbf{r}_1$ and $\mathbf{r}_2$, let's look at an "interesting" matrix multiplication

$$\mathbf{R}^T \mathbf{R} = \begin{bmatrix} \mathbf{r}_1^T \\ \mathbf{r}_2^T \end{bmatrix} @ \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1^T @ \mathbf{r}_1 & \mathbf{r}_1^T @ \mathbf{r}_2 \\ \mathbf{r}_2^T @ \mathbf{r}_1 & \mathbf{r}_2^T @ \mathbf{r}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

We call $\mathbf{R}^T \mathbf{R}$ Gram matrix.

On the diagonal of the Gram matrix $\mathbf{R}^T \mathbf{R}$, we see the inner product of each column vector with itself, which equals the square of that column vector's L^2 norm.

Meanwhile, $\mathbf{R}^T \mathbf{R}$'s off-diagonal entries correspond to the pairwise inner products of $\mathbf{R}$'s column vectors.



The book will cover the Gram matrix in detail later.

Successive Rotation

If we perform two counterclockwise rotations about the origin, first by α and then by β , the total rotation matrix is

$$\mathbf{R}(\beta)\mathbf{R}(\alpha) = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

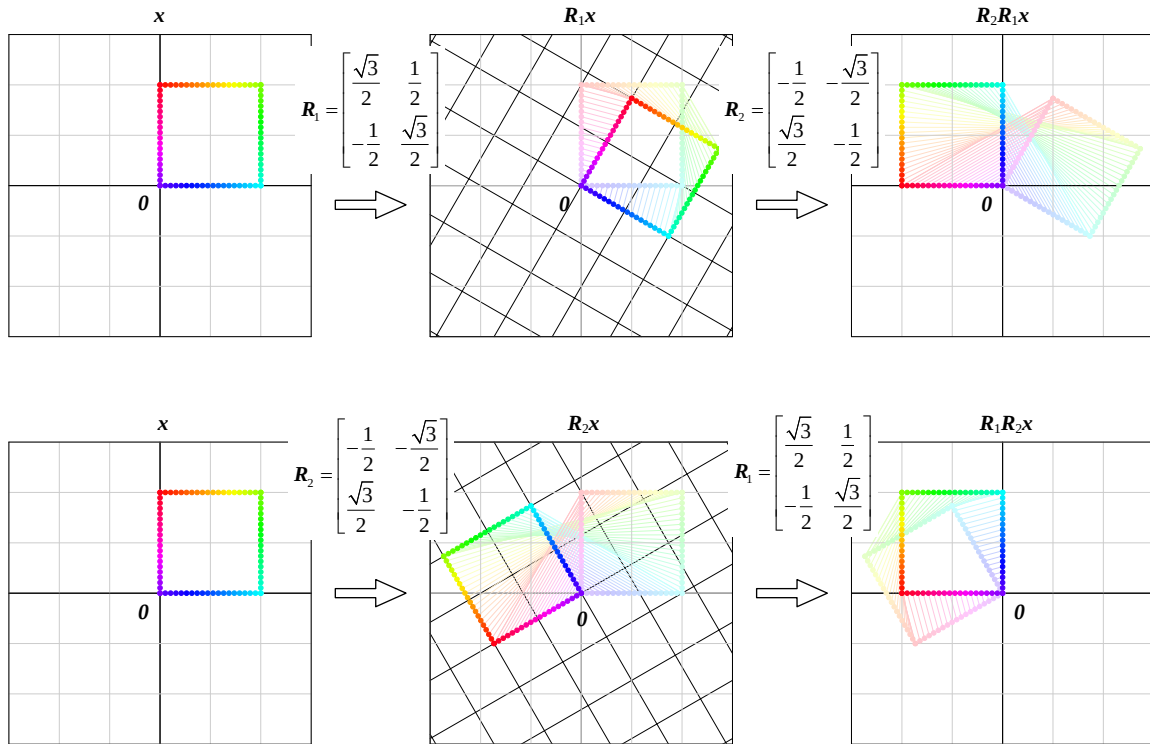
Using trigonometric identities to simplify the expression above gives

$$\mathbf{R}(\beta)\mathbf{R}(\alpha) = \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

Conversely, if we rotate first by β and then by α , the total rotation matrix is

$$\mathbf{R}(\alpha)\mathbf{R}(\beta) = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} = \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

This shows that rotating the plane about the origin is additive — multiple rotations are equivalent to a single rotation by the sum of the angles. This is also a special case where matrix multiplication satisfies the commutative law.

Figure 4. The Product of Two 2×2 Rotation Matrices Satisfies the Commutative Law

The book mentioned earlier that matrix multiplication generally does not satisfy the commutative law, that is, $AB \neq BA$.

Comparing Figure 5 and Figure 6, we can see that scaling first and then rotating gives a different result from rotating first and then scaling.

⚠ As we've repeatedly emphasized, for the matrix multiplication $ABCx$, the order in which it acts on the column vector x goes from left to right, that is C, B, A .

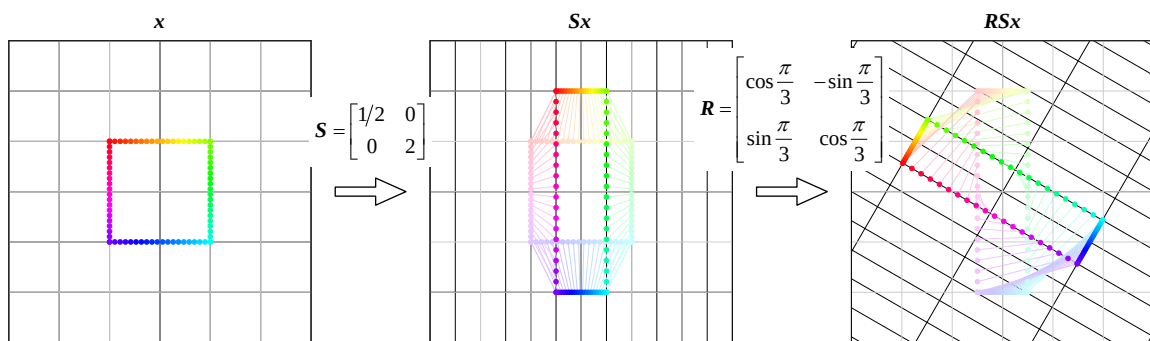


Figure 5. Scale First, Then Rotate

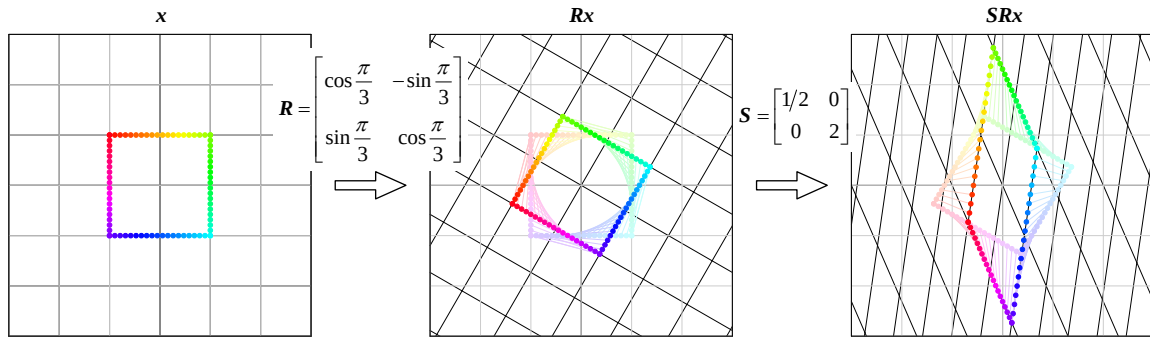


Figure 6. Rotate First, Then Scale

3D Rotation

As shown in Figure 7, in 3D space, a geometric object can independently rotate about the x_1 , x_2 , and x_3 axes.

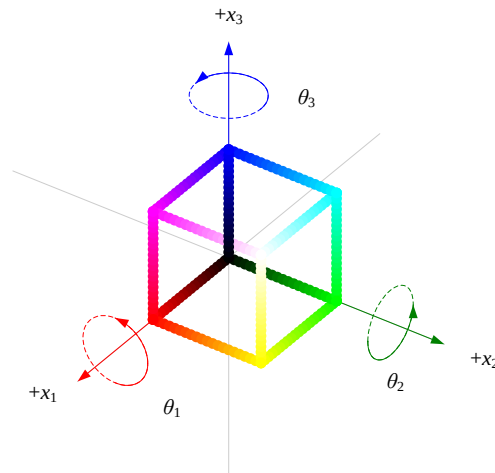


Figure 7. A Geometric Object Rotating Independently About Three Axes

In 3D space, the positive and negative directions of rotation about each axis follow the right-hand rule. Point the right thumb along the positive direction of the rotation axis; the direction the other four fingers curl indicates the positive rotation direction (counterclockwise). The direction opposite to positive rotation is negative rotation (clockwise).

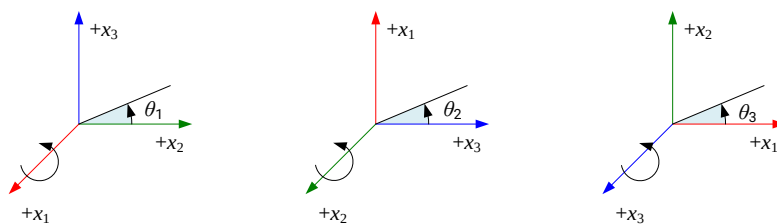


Figure 8. Three Rotation Angles

Rotation About the x_1 Axis

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

The transformation matrix for rotating counterclockwise about the x_1 axis by angle θ_1 is

$$\mathbf{R}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix}$$

As shown in Figure 9, after the 3D column vector $\mathbf{x}$ is rotated counterclockwise about the x_1 axis by angle θ_1 , its new coordinates are

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix}}_{\mathbf{R}_1} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding the expression above, we get three equations

$$\begin{cases} z_1 = x_1 \\ z_2 = x_2 \cdot \cos \theta_1 - x_3 \cdot \sin \theta_1 \\ z_3 = x_2 \cdot \sin \theta_1 + x_3 \cdot \cos \theta_1 \end{cases}$$

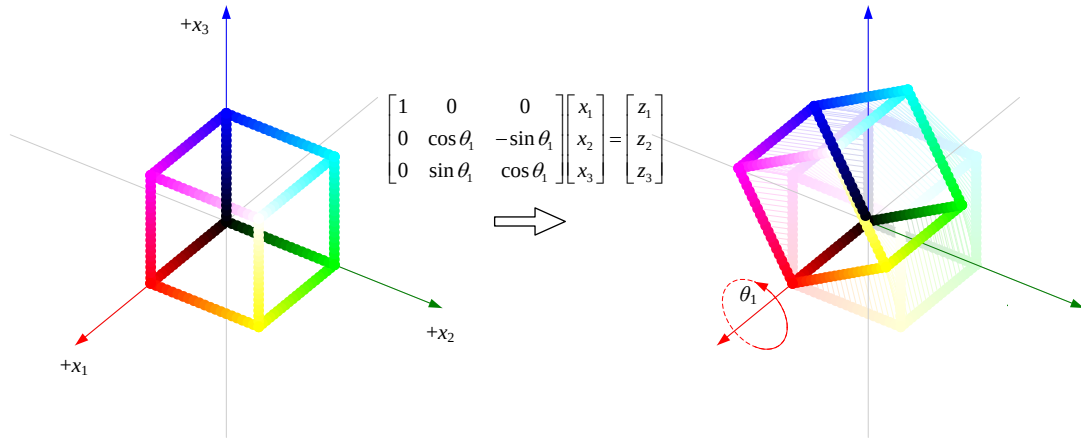


Figure 9. Rotation About the x_1 Axis, 3D Space

$\mathbf{R}_1$'s inverse matrix is

$$\mathbf{R}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & \sin \theta_1 \\ 0 & -\sin \theta_1 & \cos \theta_1 \end{bmatrix}$$

Just like the 2D rotation matrix, $\mathbf{R}_1$'s inverse corresponds to the geometric operation of rotating clockwise about the x_1 axis by angle θ_1 .

Computing $\mathbf{R}_1$'s Gram matrix

$$\mathbf{R}_1^T \mathbf{R}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & \sin \theta_1 \\ 0 & -\sin \theta_1 & \cos \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Conversely

$$\mathbf{R}_1 \mathbf{R}_1^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & \sin \theta_1 \\ 0 & -\sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This means $\mathbf{R}_1$ is also an orthogonal matrix. In 3D space, $\mathbf{R}_1$ is an orthonormal basis.

Now let's review the Laplace expansion covered earlier in the book to compute 's determinant.

Let's expand along 's first row, since this row has the most zeros and requires the least computation.

$$\det(\mathbf{R}_1) = 1 \times \begin{vmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{vmatrix} = 1$$

This means that during the rotation, the geometric object's volume doesn't change. This is fully consistent with what we observed in Figure 9.

Rotation About the x_2 and x_3 Axes

The transformation matrix for rotating counterclockwise about the x_2 axis by angle θ_2 is

$$\mathbf{R}_2 = \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix}$$

After rotating counterclockwise about the x_2 axis by angle θ_2 , the coordinates are

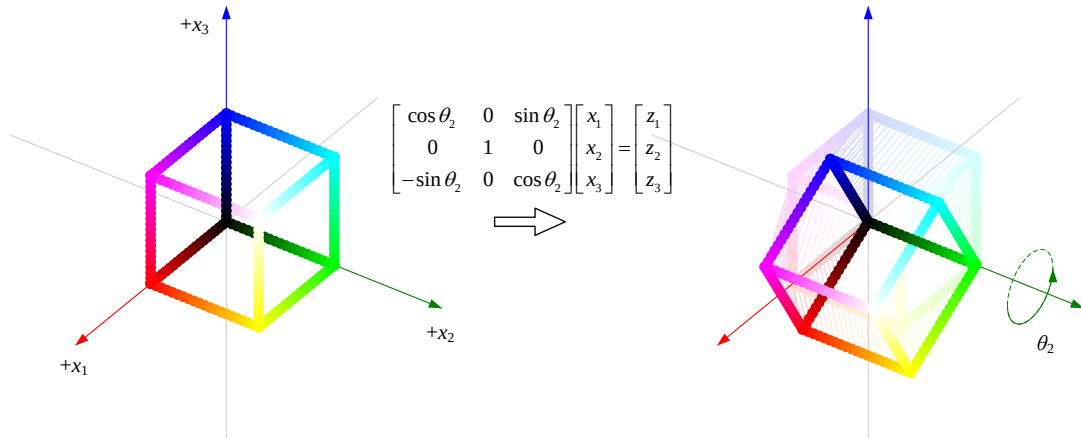
$$\underbrace{\begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix}}_{\mathbf{R}_2} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding this, we get three equations

$$\begin{cases} z_1 = x_1 \cdot \cos \theta_2 + x_3 \cdot \sin \theta_2 \\ z_2 = x_2 \\ z_3 = -x_1 \cdot \sin \theta_2 + x_3 \cdot \cos \theta_2 \end{cases}$$



Please compute the inverse of the matrix , verify that it's an orthogonal matrix, and then use Laplace expansion to compute its determinant.

Figure 10. Rotation About the x_2 Axis, 3D Space

The transformation matrix for rotating counterclockwise about the x_3 axis by angle θ_3 is

$$\mathbf{R}_3 = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

After rotating counterclockwise about the x_3 axis by angle θ_3 , the coordinates are

$$\underbrace{\begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{R}_3} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding this, we get three equations

$$\begin{cases} z_1 = x_1 \cdot \cos \theta_3 - x_2 \cdot \sin \theta_3 \\ z_2 = x_1 \cdot \sin \theta_3 + x_2 \cdot \cos \theta_3 \\ z_3 = x_3 \end{cases}$$

? Likewise, please compute the inverse of the matrix, verify that it's an orthogonal matrix, and then use Laplace expansion to compute its determinant.

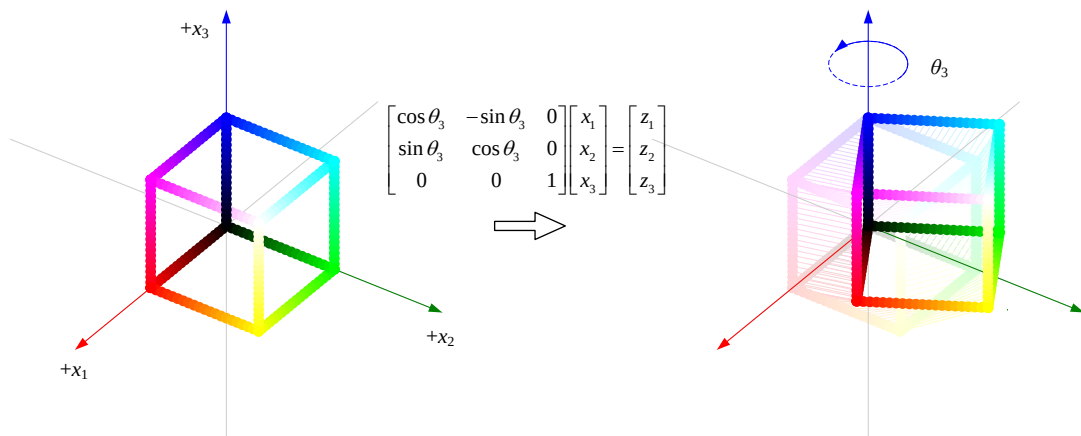


Figure 11. Rotation About the x_3 Axis, 3D Space

 Note that in 3D rotation, the order of rotations about different axes cannot be swapped; only rotations about the same axis can have their order swapped — this is similar to planar rotation.

Visualizing 3D Axis Rotation

Code 1 is a custom function for visualizing 3D axis rotation.

-  Simply assigns the original cube's vertex data to a variable `colors`, used later to set each point's color. The unit cube's vertex coordinates double as the RGB color space — each vertex's coordinates serve as its color code.
-  Defines a list that enumerates the cube's 12 edges, each edge given by the indices of its two endpoints. For example, `[0, 1]` means the edge connecting point 0 and point 1. There are 8 points in total, forming 12 edges.
-  Uses a for loop to draw all the edges of the cube in its original state. `original_cube[edge]` retrieves the edge's two vertices, and then `zip(*...)` extracts their x, y, and z coordinates separately. The `*` in front is the unpacking operator, telling the plot function to “treat these as separate x, y, z coordinates.” `color="gray"` sets the line to gray, and `linestyle="dotted"` sets it to a dotted line.
-  Similar to , draws the shape after the geometric transformation.
-  Uses `scatter()` to draw the original cube's vertices, with a different color set for each vertex.
-  Similar to , uses `scatter()` to draw the transformed cube's vertices

Code 1. Custom Visualization Function |  LA_08_03_01.ipynb

```

## 可视化函数
def plot_cube(ax, original_cube, transformed_cube):
    colors = original_cube

    # 定义每个卦限的框线顶点对
    edges = [[0, 1], [1, 2], [2, 3], [3, 0],
             [0, 4], [1, 5], [2, 6], [3, 7],
             [4, 5], [5, 6], [6, 7], [7, 4]]

    # 绘制原始立方体框线
    for edge in edges:
        ax.plot(*zip(*original_cube[edge]),
                color="gray", linestyle="dotted")

    # 绘制变换后的立方体框线
    for edge in edges:
        ax.plot(*zip(*transformed_cube[edge]),
                color="black", linestyle="-")

    # 绘制顶点
    ax.scatter(original_cube[:,0], original_cube[:,1],
               original_cube[:,2],
               facecolor=colors, s=50, marker='o', edgecolor='k')

    ax.scatter(transformed_cube[:,0], transformed_cube[:,1],
               transformed_cube[:,2],
               facecolor=colors, s=50, marker='o', edgecolor='k')

    ax.set_xlim(-2, 2); ax.set_ylim(-2, 2); ax.set_zlim(-2, 2)
    ax.axis(False); ax.set_box_aspect([1, 1, 1])
    ax.view_init(elev=30, azimuth=30)
    ax.set_proj_type('ortho')
    ax.plot((-2,2),(0,0),(0,0),color = 'k')
    ax.plot((0,0),(-2,2),(0,0),color = 'k')
    ax.plot((0,0),(0,0),(-2,2),color = 'k')

```

Code 2 calls Code 1's custom functions to draw Figure 12. Code 2 is quite simple — please annotate it line by line on your own.

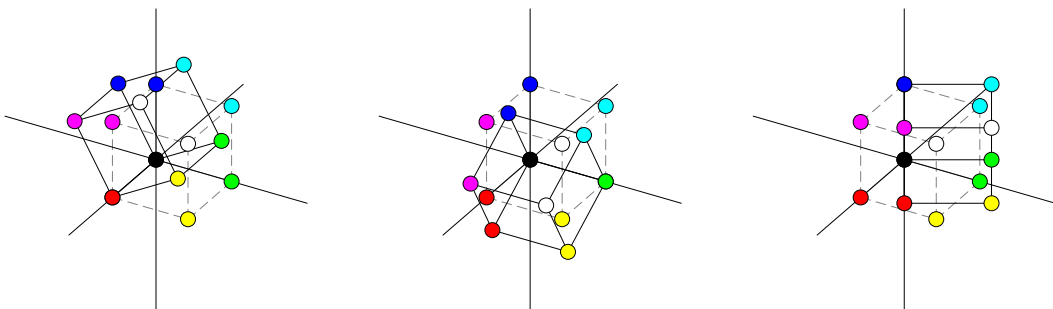


Figure 12. A 3D Axis Rotation Example

Code 2. 3D Axis Rotation | LA_08_03_01.ipynb

```

## 定义单位立方体的 8 个顶点坐标
a cube = np.array([[0, 0, 0], [1, 0, 0], [1, 1, 0], [0, 1, 0],
                  [0, 0, 1], [1, 0, 1], [1, 1, 1], [0, 1, 1]])

## 绕 x1 轴旋转
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111, projection='3d')

b theta_1 = np.pi / 6
R_x1 = np.array([[1, 0, 0],
                [0, np.cos(theta_1), -np.sin(theta_1)],
                [0, np.sin(theta_1), np.cos(theta_1)]])

rotated_cube = cube @ R_x1.T
plot_cube(ax, cube, rotated_cube)

## 绕 x2 轴旋转
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111, projection='3d')

c theta_2 = np.pi / 6
R_x2 = np.array([[np.cos(theta_2), 0, np.sin(theta_2)],
                [0, 1, 0],
                [-np.sin(theta_2), 0, np.cos(theta_2)]])

rotated_cube = cube @ R_x2.T
plot_cube(ax, cube, rotated_cube)

## 绕 x3 轴旋转
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111, projection='3d')

d theta_3 = np.pi / 6
R_x3 = np.array([[np.cos(theta_3), -np.sin(theta_3), 0],
                [np.sin(theta_3), np.cos(theta_3), 0],
                [0, 0, 1]])

rotated_cube = cube @ R_x3.T
plot_cube(ax, cube, rotated_cube)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Write Python code that uses matrix multiplication to rotate the point (3, 4) counterclockwise about the origin by 90° , and compute the coordinates after rotation.

Q2. Prove that the inverse of the 2D rotation matrix is its transpose.

Q3. For planar rotation, first rotate by 30° and then by 60° ; compute each rotation matrix separately, and then compute the composite transformation matrix.

Q4. Write a Python function `rotate_point(x1, x2, theta)` that takes a point $(x1, x2)$ and an angle θ (in radians) as input, and returns the coordinates after rotating about the origin in the plane.

Q5. Given a set of planar point coordinates `points = [(1, 0), (0, 1), (-1, 0), (0, -1), (0, 0)]`, write Python code to rotate them counterclockwise by 45° and output the new coordinates.

Q6. Write Python code that uses `matplotlib` to draw a unit square, then draws the unit square again after rotating it counterclockwise about the origin by 60° .

Q7. Write code to compare whether the following two 3D rotations, performed in different orders, give the same result

- First rotate 90° about x_1 , then rotate 90° about the x_2 axis;
- First rotate 90° about x_2 , then rotate 90° about the x_1 axis;

Q8. Learn what Rodrigues' rotation is.

Q9. Learn what a unit quaternion is, and how it can be used to perform rotation in 3D space.

Q10. Learn about the gimbal lock problem.

Q11. Learn about complex numbers, and how they relate to rotation.